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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Basic Electrical Engineering (New)

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

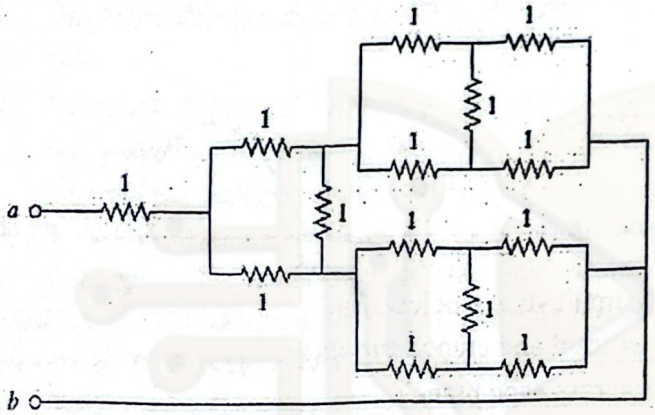
Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

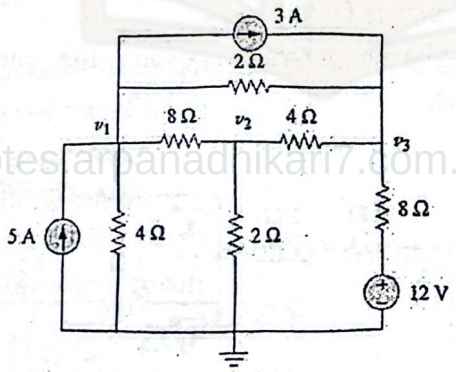
Attempt all the questions.

1. a) Classify types of consumers based upon their electricity consumption pattern. Also draw the single line diagram of a general power system. 7

   b) Find equivalent resistance at a-b R_{ab} in the figure. Assume each element is 1Ω. 8



2. a) Using nodal analysis to find v_1, v_2, and v_3 in the following circuit. 7



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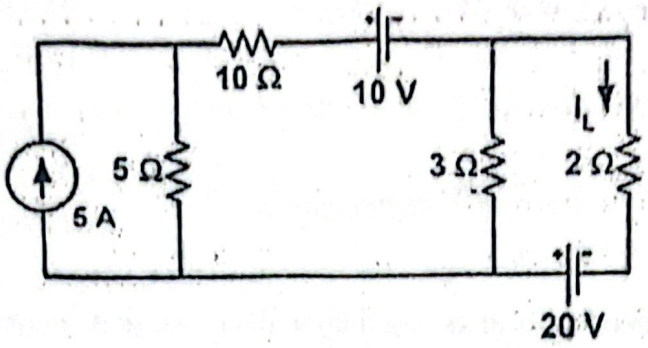
OR

What are active and passive elements, unilateral and bilateral element? What is star delta transformation and writes its role in circuit analysis?

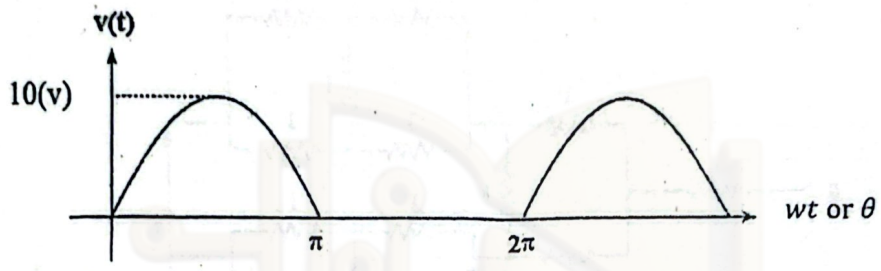
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b) State Superposition Theorem power theorem. Find voltage drop through 2-ohm resistor using Norton's Theorem in given circuit. 8



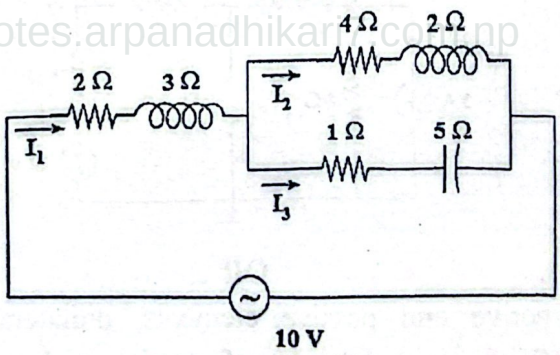
3. a) The waveform given is the half wave rectified sine wave. Find the average, RMS value and form factor. Also find the amount of average power dissipated in 10 Ω. 8



b) For the electric circuit shown below, find: 7

i. supply current and branch currents
ii. voltage across each branch
iii. Active, reactive and apparent power of entire circuit
iv. overall power factor
v. Draw phasor diagram showing the currents and supply voltages.

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4. a) With the help of series resonance curve, explain resonance in series RLC circuit. 7

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drop

	b)	Three identical coils, each of resistance 10 Ω and inductance 42 mH are connected a) in star b) in delta to a 415 V, 50 Hz, 3-φ supply. Determine: i. line and phase currents ii. Line and phase voltages iii. the total power dissipated in each case and compare them.	8
5.	a)	Explain two wattmeter method for three phase power measurement. Also, how it is different from one wattmeter and three wattmeter method.	7
	b)	A 5 kVA, 200/1000 V, 50 Hz, single phase transformer gave the following test results: OC test: 200V, 1.2 A, 90 W SC test: 50 V, 5A, 110 W i. Calculate the parameters of the equivalent circuit referred to LV side. ii. Calculate the output secondary voltage when delivering 3 kW at 0.8 power factor lagging, the input voltage being 200 V. Find also the percentage voltage regulation.	8
6.	a)	A 250 V DC shunt motor takes 8 A line current on no load and runs at 1000 rpm. The resistance of the field winding and armature winding are 200 Ohm and 0.8 Ohm respectively. If the full load line current is 20 A, calculate the full load speed assuming constant air gap flux.	7
	b)	Explain the construction of DC generator with working principle.	8
		OR	
		Explain the importance of transformer in power transmission. Derive the EMF equation of transformer.	
7.		Write short notes on: **(Any two)** a) Speed Control of DC Motor b) Color coding of resistors c) Power in purely inductive circuit	2×5

BEE-2024 SP

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